

Section 13.1.

1. Show that $p(x) = x^3 + 9x + 6$ is irreducible in $\mathbb{Q}[x]$.

And, for a root θ of $p(x)$, find $(1+\theta)^{-1}$ in $\mathbb{Q}(\theta)$.

Proof. Since $9, 6 \in (3)$ but $6 \notin (3^2)$, so the Eisenstein Criterion gives p is irreducible.

Let θ be the root of $p(x)$. By the Division Algorithm,

$$x^3 + 9x + 6 = (x+1)(x^2 - x + 10) - 4$$

$$\text{Thus, } 4 = (\theta+1)(\theta^2 - \theta + 10) - (\theta^3 + 9\theta + 6) = (\theta+1)(\theta^2 - \theta + 10), \text{ or}$$

$$(\theta+1)^{-1} = \frac{1}{4} \cdot (\theta^2 - \theta + 10).$$

2. Show that $x^3 - 2x - 2$ is irreducible over $\mathbb{Q}$ and let θ be a root.

Compute $(1+\theta)(1+\theta+\theta^2)$ and $(1+\theta)(1+\theta+\theta^2)^{-1}$.

$$\text{Proof. } (1+\theta)(1+\theta+\theta^2) = 1 + \theta + \theta^2 + \theta + \theta^2 + \theta^3 = \theta^3 + 2\theta^2 + 2\theta + 1 \\ = 2\theta^2 + 2\theta + 1 + 2\theta + 2 = 2\theta^2 + 4\theta + 3.$$

And,

$$(x^3 - 2x - 2) = (x^2 + x + 1)(x - 1) + (-2x - 1)$$

$$(x^2 + x + 1) = (-2x - 1)\left(-\frac{1}{2}x - \frac{1}{4}\right) + \frac{3}{4}$$

$$\Rightarrow (x^2 + x + 1) = (x^3 - 2x - 2)\left(-\frac{1}{2}x - \frac{1}{4}\right) - (x^2 + x + 1)(x - 1)\left(-\frac{1}{2}x - \frac{1}{4}\right) + \frac{3}{4}$$

$$\text{So, } (\theta^2 + \theta + 1) = 0 - (\theta^2 + \theta + 1)(\theta - 1)\left(-\frac{1}{2}\theta - \frac{1}{4}\right) + \frac{3}{4},$$

$$(\theta^2 + \theta + 1)\left(-\frac{1}{2}\theta^2 + \frac{1}{4}\theta + \frac{5}{4}\right) = \frac{3}{4},$$

$$\text{So, } (\theta^2 + \theta + 1) = \frac{4}{3}\left(-\frac{1}{2}\theta^2 + \frac{1}{4}\theta + \frac{5}{4}\right)$$

$$= \frac{1}{3}(-2\theta^2 + \theta + 5).$$

#13.1.3. Show that x^3+x+1 is irreducible over $\mathbb{F}_2$.

And let θ be a root of the polynomial. Compute θ^n in $\mathbb{F}_2(\theta)$.

Proof. $0+0+1 \neq 0$, $1+1+1 \neq 0$, So, x^3+x+1 has no root in $\mathbb{F}_2$,
and it has degree 3, So irreducible.

And, $\theta = \theta$, $\theta^2 = \theta^2$, $\theta^3 = -\theta - 1 = \theta + 1$ (in $\mathbb{F}_2$)

$$\theta^4 = \theta^2 + \theta$$

$$\theta^5 = \theta^3 + \theta^2 = \theta^2 + \theta + 1$$

$$\theta^6 = \theta^3 + \theta^2 + \theta = \theta^2 + 2\theta + 1 = \theta^2 + 1$$

$$\theta^7 = \theta^3 + \theta = 2\theta + 1 = 1.$$

Thus, $(\mathbb{F}_2[x])/(x^3+x+1)$ is a field, order of 8.

#13.1.5. Let α is a rational root of a monic polynomial in $\mathbb{Z}[x]$.

Prove that α is integer.

Proof. Let $\alpha = r/s$, where r and s are relatively prime integers. Then,

$$\frac{r^n}{s^n} + a_{n-1} \cdot \frac{r^{n-1}}{s^{n-1}} + \dots + a_1 \cdot \frac{r}{s} + a_0 = 0.$$

$$\Rightarrow r^n + a_{n-1} \cdot s r^{n-1} + \dots + a_1 s^{n-1} r + a_0 s^n = 0.$$

$$\Rightarrow r^n = s \cdot (-a_{n-1} r^{n-1} - a_{n-2} s r^{n-2} - \dots - a_0 s^{n-1})$$

That is, s divides r^n . But, since $(s, r) = 1$, it is contradiction.

#13.1.7. Prove that x^3-nx+2 is irreducible, except $n = -1$ or 3 or 5 .

Proof.

Sec 13.2.

3. $x = 1+i$. $(x-1)^2 = -1$, so $x^2 - 2x + 2 = 0$.

It is minimal, because $1+i \notin \mathbb{Q}$.

4. Determine the degree over $\mathbb{Q}$ of $2+\sqrt{3}$ and $1+\sqrt[3]{2}+\sqrt[3]{4}$.

Proof. $2+\sqrt{3} \in \mathbb{Q}(\sqrt{3})$, so $\mathbb{Q}(2+\sqrt{3}) \subseteq \mathbb{Q}(\sqrt{3})$, and since $\mathbb{Q} \neq \mathbb{Q}(2+\sqrt{3})$, and $[\mathbb{Q}(\sqrt{3}) : \mathbb{Q}] = 2$, so $\deg 2+\sqrt{3} = 2$.

And, since $1+\sqrt[3]{2}+\sqrt[3]{4} \in \mathbb{Q}(\sqrt[3]{4})$, again, similarly, $\deg 1+\sqrt[3]{2}+\sqrt[3]{4} = 3$.

5. Let $F = \mathbb{Q}(i)$. Prove x^3-2 and x^3-3 are irreducible over F .

Proof. Since both are degree 3, the one is irreducible iff one has a root.

Let α be a root in $\mathbb{Q}(i)$ of x^3-2 . Then, $\mathbb{Q}(\alpha)$ has degree 3,

since x^3-2 is irreducible over $\mathbb{Q}$, by the Eisenstein criterion.

But, $[\mathbb{Q}(i) : \mathbb{Q}] = 2$, this is contradiction to $\mathbb{Q}(\alpha) \subseteq \mathbb{Q}(i)$.

Similarly x^3-3 is irreducible.

7. Prove that $\mathbb{Q}(\sqrt{2}+\sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

Proof. Clearly, $\sqrt{2}+\sqrt{3} \in \mathbb{Q}(\sqrt{2}, \sqrt{3})$, so $\subseteq$ is proved.

Conversely, $\sqrt{2}+\sqrt{3} = x \Rightarrow 2 = x^2 - 2\sqrt{3}x + 3$

$$\Rightarrow 2\sqrt{3}x = x^2 + 1$$

$$\Rightarrow 12x^2 = x^4 + 2x^2 + 1$$

$$\Rightarrow x^4 - 10x^2 + 1 = 0.$$

It is irreducible, because $(x^2)^2 - 10(x^2) + 1 = 0$ has no root in $\mathbb{Q}$.

So, $\deg_{\mathbb{Q}} \sqrt{2}+\sqrt{3} = 4$. Meanwhile, since $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$, so these are same.

#8, 9. Biquadratic.

Let F be a field of characteristic $\neq 2$.

Let $D_1, D_2 \in F$, neither of which is a square in F .

Prove that: If $D_1 D_2$ is not a square in F , then $F(\sqrt{D_1}, \sqrt{D_2})$ is degree 4.

If $D_1 D_2$ is a square in F , then $F(\sqrt{D_1}, \sqrt{D_2})$ is degree 2.

Proof. First, $[F(\sqrt{D_1}) : F] = 2$ is clear.

So, enough to show that $[F(\sqrt{D_1}, \sqrt{D_2}) : F(\sqrt{D_1})] = 2$

If these are same, then $\sqrt{D_2} \in F(\sqrt{D_1})$, so $\sqrt{D_2} = a_1 \sqrt{D_1} + a_2$ for some $a_1, a_2 \in F$.

Now, $D_2 - a_2^2 D_1 = a_2(\sqrt{D_2} + a_1 \sqrt{D_1}) = a_2(2a_1 \sqrt{D_1} + a_2) = 2a_1 a_2 \sqrt{D_1} + a_2^2$.

$\Rightarrow 2a_1 a_2 \sqrt{D_1} = D_2 - a_2^2 D_1 + a_2^2$.

$2 \neq 0$, since $\text{ch}(F) \neq 2$, and $a_1 \neq 0$, because $a_1 = 0$, then $\sqrt{D_2} = a_2 \in F$, and $a_2 \neq 0$; if $a_2 = 0$, then $\sqrt{D_2} = a_1 \sqrt{D_1} \Rightarrow \sqrt{D_2} = a_1 \in F$.

So, $\sqrt{D_1} = (2a_1 a_2)^{-1}(D_2 - a_2^2 D_1 + a_2^2) \in F$, contradiction.

If $\sqrt{D_1 D_2} \in F$, then $\sqrt{D_1} \in F(\sqrt{D_2})$, so prove.

Let $a, b \in F$ and b is not a square in F . Then,

$\sqrt{a+b} = \sqrt{m} + \sqrt{n}$ for some $m, n \in F$ iff $\sqrt{a^2 - b} \in F$.

Proof. Observe that if $\sqrt{a+b} = \sqrt{m} + \sqrt{n}$ for some $m, n \in F$, then

$\sqrt{a+b} = \sqrt{m} + \sqrt{n} \Rightarrow a+b = m+n+2\sqrt{mn} \Rightarrow a-(m+n) = 2\sqrt{mn} - \sqrt{b} \in F$,

Then $2\sqrt{mn} - \sqrt{b}$ must be zero; let $2\sqrt{mn} - \sqrt{b} = k \in F$. Then

$4mn = k^2 + b + 2k\sqrt{b}$. If $k \neq 0$, then $\sqrt{b} = \frac{1}{2k} \cdot (k^2 - b + 4mn)$, so $\sqrt{b} \in F$.

(And $2 \neq 0$, by $\text{ch}(F) \neq 2$)

So $2\sqrt{mn} = \sqrt{b}$ and $a = m+n$, so $a^2 - b = (m+n)^2$.

Conversely, $m = \frac{1}{2}(a + \sqrt{a^2 - b})$ and $n = \frac{1}{2}(a - \sqrt{a^2 - b})$, then we obtained the left.

Now, $\mathbb{Q}(\sqrt{a+b})$ ($a, b \in \mathbb{Q}$) has degree 4 iff $a^2 - b$ is not square in $\mathbb{Q}$.

So, $\mathbb{Q}(\sqrt{3+2\sqrt{2}})$ has degree 2 $9-8=1$.

More specifically, since $3^2 - 8 = 1$, $\sqrt{a^2 - b} = 1$. So, Take $m = \frac{1}{2}(3+1) = 2$, $n = 1$,

$\sqrt{3+2\sqrt{2}} = \sqrt{2+1}$

13. Suppose $F = \mathbb{Q}(\alpha_1, \dots, \alpha_n)$ where $\alpha_i^2 \in \mathbb{Q}$.

prove $\sqrt[3]{2} \notin F$.

Proof. If $\sqrt[3]{2} \in F$, then $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{2}) \subseteq F$.

But, since $F = (\mathbb{Q}(\alpha_1)(\alpha_2) \dots (\alpha_n))$, so it has degree 2^m for some $1 \leq m \leq n$,

And $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$, so $3 \mid 2^n$, contradiction.

14. If $[F(\alpha) : F]$ is odd, then $F(\alpha) = F(\alpha^2)$.

Proof. Clearly $\alpha^2 \in F(\alpha)$, so $F(\alpha^2) \subseteq F(\alpha)$. Thus,

$$[F(\alpha) : F(\alpha^2)] = 1 \text{ or } 2.$$

If the degree is 2, then $[F(\alpha) : F] = [F(\alpha) : F(\alpha^2)] [F(\alpha^2) : F]$ cannot be odd.

#13.2.18.

Let K be a field and $K(x)$ be the field of rational functions in x .

Let $t = P(x)/g(x) \in K(x)$ with $P(x)$ and $g(x)$ relatively prime, $g(x) \neq 0$.

a). Show that the polynomial $P(X) - t \cdot g(X) \in K(t)[X]$ is irreducible over $K(t)$, and has x as a root.

Proof. Since $P(x) - t \cdot g(x) = P(x) - P(x) = 0$, so $P(X) - t \cdot g(X)$ has x as a root.

And, By the Gauss Lemma, $P(X) - t \cdot g(X)$ is irreducible over $K(t)$ iff over $K[t]$.

because the field of fraction of $K[t]$ is $K(t)$, and the gcd of coefficients is 1, by the hypothesis. And, since $K[t][X] = K[X][t]$ and clearly $P(X) - t \cdot g(X)$ is irreducible over $K[X]$, if it is reducible, then $P(X)$ and $g(X)$ are not relatively prime.

So, $P(X) - t \cdot g(X)$ is irreducible over $K[X]$, so is over $K[t]$, so is $K(t)$.

b). Show that $P(X) - t \cdot g(X) \in K(t)[X]$ has degree of $\max\{\deg P(x), \deg g(x)\}$.

Proof. There are three possible cases,

1) $\deg P > \deg g$ 2) $\deg P = \deg g$ 3) $\deg P < \deg g$

If 1), then clearly degree of $P(X) - t g(X)$ is $\deg P$.

If 3), similarly clear but: let $a(t) \in K[t]$ be the leading coefficient of $g(X)$.

$t \cdot a(t)$ cannot be zero, because if it is zero, then t is algebraic over K .

Similarly to 2), the degree is preserved in the case of 3).

c). Show that $[K(x):K(t)] = [K(x):K(P(x)/g(x))] = \max(\deg P, \deg g)$.

Proof. As a result of a) and b), $P(X) - t \cdot g(X)$ is irreducible over $K(t)$ and has x as a root, so proved.

#13.2.19~21.

Let K/F be an Extension of degree n .

1). Given $\alpha \in K$, α acting by left multiplication on K is an F -linear transformation of K .

Proof. Let $\alpha \in K$ be fixed. Then, $T_\alpha(x) = \alpha x$ ($x \in K$) satisfies

$$\alpha(cx + y) = \alpha cx + \alpha y = cT_\alpha(x) + T_\alpha(y) \quad \text{for all } x, y \in K \text{ and } c \in F.$$

So, it is linear.

2) K is isomorphic to a subfield of the ring $M_{n \times n}(F)$.

Proof. Given α , $T_\alpha: K \rightarrow K: x \mapsto \alpha x$ is a linear map over F . Then,

$$K \rightarrow \mathcal{L}(K, K): \alpha \mapsto T_\alpha$$

is an injective Homomorphism, so isomorphic. More specifically,

Let $\beta = \{d_1, \dots, d_n\}$ be the basis for K over F . Then,

$$[T_\alpha]_\beta = \begin{pmatrix} c_{11} & \dots & c_{1n} \\ c_{21} & & i \\ \vdots & & \\ c_{n1} & \dots & c_{nn} \end{pmatrix} \quad \text{where } \alpha \cdot d_i = c_{i1} \cdot d_1 + \dots + c_{in} \cdot d_n, \quad c_{ij} \in F.$$

$$\text{So, } \{[T_\alpha]_\beta \mid \alpha \in K\} \leq M_{n \times n}(F).$$

3). Show that characteristic polynomial of T_α has a root as α .

Proof. Given $K \in K$,

$T_\alpha(K) = \alpha K$, so α is an eigenvalue of T_α , so it is a root of the characteristic.

In specifically, observe $\mathbb{Q}(\sqrt[3]{2}) = \text{span}_{\mathbb{Q}} \{1, \sqrt[3]{2}, \sqrt[3]{4}\}$.

Sec 14.2. Galois.

Sec 14.3 Finite Field.

1. Factor $x^8 - x$ in $\mathbb{Z}[x]$ and $\mathbb{F}_2[x]$.

$$\text{Proof } x^8 - x = x \cdot (x^7 - 1) = x(x-1)(x^6 + x^5 + x^4 + x^3 + x^2 + x + 1)$$

In $\mathbb{Z}[x]$, $x^6 + \dots + 1$ is irreducible, because it is irreducible in $\mathbb{Q}$ and using the Gauss Lemma. But, in $\mathbb{F}_2$,

$$x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = (x^3 + x^2 + 1)(x^3 + x + 1)$$

$$\text{So } x^8 - x = x(x-1)(x^3 + x^2 + 1)(x^3 + x + 1).$$

2.

18. Let $\alpha \in \mathbb{F}_p$ be a non-zero, and $p(x) = x^p - x - \alpha \in \mathbb{F}_p[x]$.
 Determine the splitting field of $p(x)$, and find the Galois group.
 Proof. First, $p(x)$ is separable, because the formal derivative

$$D_x p(x) = px^{p-1} - 1 = -1$$

is relatively prime with $p(x)$.

Let $\alpha \in \overline{\mathbb{F}_p}$ be the root of $p(x)$. That is,

$$\alpha^p - \alpha - \alpha = 0.$$

Then, $(\alpha+1)^p - (\alpha+1) - \alpha = \alpha^p + 1 - (\alpha+1) - \alpha = \alpha^p - \alpha - \alpha = 0$.

Thus, inductively, $\alpha, \alpha+1, \alpha+2, \dots, \alpha+p-1$ are all roots of $p(x)$.

And, since $\alpha \neq 0$, $p(x)$ cannot have a root in $\mathbb{F}_p$.

Now, suppose that $p(x)$ is reducible by $p(x) = f(x)h(x)$. Here, $f, h \in \mathbb{F}_p[x]$.

Since $p(x)$ is separable, $\alpha, \alpha+1, \dots, \alpha+p-1$ are all distinct, so

$$f(x) = \prod_{i \in I} (x - (\alpha+i))$$

where $I \subset \{0, 1, \dots, p-1\}$. Then, for $n = |I|$,

$$f(x) = x^n - \left(\sum_{i \in I} \alpha+i \right) x^{n-1} + \dots \in \mathbb{F}_p[x]$$

Since $\sum_{i \in I} \alpha+i = n\alpha + \sum_{i \in I} i \in \mathbb{F}_p$ and $\sum_{i \in I} i \in \mathbb{F}_p$,

we obtain that $n\alpha \in \mathbb{F}_p$, so $n = 0$ or p .

Either $n = 0$ or $n = p$, it is contradiction to p is reducible.

Now, the splitting field of p is $\mathbb{F}_{p^p}$ and $\text{Gal}(\mathbb{F}_{p^p}/\mathbb{F}_p) = \mathbb{Z}_p$, cyclic.